

Qiskit v2.x Practice Quiz

Questions

1. What does the following circuit prepare?

```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
```

- A. $|00\rangle$
- B. $|11\rangle$
- C. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- D. $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

2. What is the purpose of transpilation?

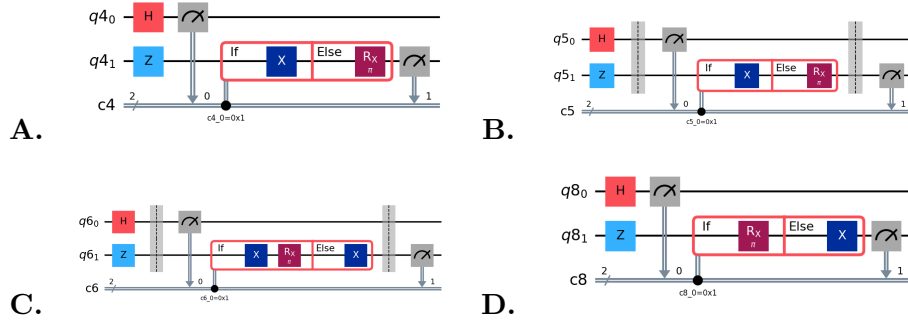
- A. To convert a circuit into backend-compatible instructions
- B. To measure all qubits automatically
- C. To create a statevector
- D. To define an observable

3. Which one of the following images is the output from the code below:

```
from qiskit import *
from math import pi

qubits = QuantumRegister(2)
clbits = ClassicalRegister(2)
qc = QuantumCircuit(qubits, clbits)
(q0, q1) = qubits
(c1, c2) = clbits

qc.h(0)
qc.z(1)
qc.barrier()
qc.measure(q0, c1)
with qc.if_test((c1,1)) as else_:
    qc.x(q1)
with else_:
    qc.rx(pi, q1)
qc.barrier()
qc.measure(q1, c2)
qc.draw(output="mpl", plot_barriers=False)
```



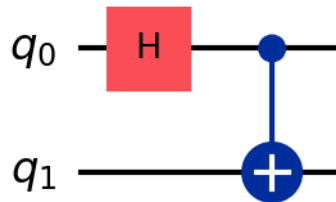
4. What does the Sampler primitive primarily return?

- A. Expectation values
- B. Measurement samples or bitstring results
- C. Only statevectors
- D. Only transpiled circuits

5. What does layout refer to in transpilation?

- A. Mapping virtual qubits to physical qubits
- B. Choosing the number of shots
- C. Drawing the circuit
- D. Choosing classical register names

6. What state does the circuit below prepare?



- A. $|00\rangle$
- B. $|11\rangle$
- C. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- D. $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

7. Which OpenQASM 3 variable declaration corresponds to Qiskit's ClassicalRegister declaration?

- A. bit
- B. clreq[]
- C. bit[]
- D. clbit[]

8. Which of these is not a valid Estimator PUB entry?
- A. a single parameterized quantum circuit
 - B. a single quantum circuit that contains no parameters
 - C. target precision
 - D. number of shots
9. What does an X gate do to $|0\rangle$?
- A. Creates $|+\rangle$
 - B. Maps it to $|1\rangle$
 - C. Measures the qubit
 - D. Adds a global phase only
10. Which object is commonly used to represent Pauli observables in Qiskit?
- A. QuantumCircuit
 - B. SparsePauliOp
 - C. ClassicalRegister
 - D. PassManager
11. Which of the following state vectors corresponds to the following OpenQASM 3 code?
- ```
OPENQASM 3;
include "stdgates.inc";

qubit [2] q;
bit [2] c;

h q[0];
cx q[1], q[0];
z q[1];
```
- A.  $\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |01\rangle$
  - B.  $\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$
  - C.  $\frac{\sqrt{2}}{2} |00\rangle - \frac{\sqrt{2}}{2} |01\rangle$
  - D.  $\frac{1}{\sqrt{2}} |00\rangle - \frac{1}{\sqrt{2}} |11\rangle$
12. Which state lies on the positive X axis of the Bloch sphere?
- A.  $|0\rangle$
  - B.  $|1\rangle$
  - C.  $|+\rangle$
  - D.  $|-\rangle$
13. Why may SWAP gates be inserted during transpilation?
- A. To reset qubits

- B. To satisfy hardware connectivity constraints
  - C. To increase shot count
  - D. To remove all noise
14. You can access your submitted jobs by providing `QiskitRuntimeService().jobs()` method which of the following? (*Select all that apply*)
- A. date jobs were created after
  - B. date jobs were created before
  - C. CRN
  - D. API key
15. Which code correctly creates a two-qubit quantum circuit?
- A. `QuantumCircuit()`
  - B. `QuantumCircuit(2)`
  - C. `Circuit(2)`
  - D. `QuantumRegister.create(2)`
16. What does increasing the number of shots usually improve?
- A. Sampling precision
  - B. Circuit depth
  - C. Qubit connectivity
  - D. Gate fidelity automatically
17. When is Sampler more appropriate than Estimator?
- A. When you need bitstring measurement outcomes
  - B. When you need expectation values of observables
  - C. When you need to draw a circuit
  - D. When you need to create a backend
18. Which gate maps  $|0\rangle$  to  $|+\rangle$ ?
- A. X
  - B. Z
  - C. H
  - D. CX
19. What is the result of applying Z to  $|1\rangle$ ?
- A.  $|0\rangle$
  - B.  $|1\rangle$
  - C.  $-|1\rangle$
  - D.  $|+\rangle$
20. What does the Bloch sphere represent?

- A. Only classical bits
- B. Single-qubit pure states
- C. Only two-qubit entangled states
- D. Classical probability distributions only

## **Answer Key**

1. S3-002: C
2. S4-001: A
3. S3-003: A
4. S5-001: B
5. S4-003: A
6. S3-004: C
7. S8-003: C
8. S6-007: D
9. S1-001: B
10. S6-002: B
11. S8-004: A
12. S2-002: C
13. S4-002: B
14. S7-007: A, B, C
15. S3-001: B
16. S7-001: A
17. S5-002: A
18. S1-002: C
19. S1-003: C
20. S2-001: B